

Phonon Frequency Spectrum Benchmarking for Interatomic Potentials: A Technical Review for the GLIM Project

1. Foundational Concepts and Scope

1.1 Phonon Frequency Spectrum as a Critical Validation Target

1.1.1 Physical significance of harmonic phonon properties in materials characterization The phonon frequency spectrum represents one of the most fundamental and stringent tests of interatomic potential accuracy, as it directly probes the **curvature of the potential energy surface (PES)** at equilibrium configurations. Unlike total energies and atomic forces—which depend on zeroth and first derivatives of the PES—phonon frequencies emerge from **second derivatives of energy with respect to atomic displacements**, making them exponentially more sensitive to subtle errors in potential parameterization ([arXiv.org](#)). This heightened sensitivity explains why phonon benchmarks have become indispensable for validating both classical empirical potentials and modern machine learning interatomic potentials (MLIPs), particularly for applications requiring accurate thermal and mechanical property predictions.

The physical significance of phonon properties extends across multiple scales. At the atomic level, phonon frequencies determine **vibrational mode energies** that govern spectroscopic signatures; at the mesoscale, they control **thermal conductivity** through phonon group velocities and scattering rates; and at the macroscale, they dictate **phase stability** via vibrational entropy contributions to free energy ([Emergent Mind](#)). For crystalline inorganic materials, the maximum phonon frequency **__max** serves as a particularly valuable scalar diagnostic, reflecting the stiffest interatomic bonds and correlating strongly with Debye temperature and high-temperature heat capacity ([arXiv.org](#)). Recent large-scale studies have demonstrated that achieving **DFT-level phonon accuracy** (errors smaller than intrinsic PBE-PBESol functional differences) is now possible with well-trained universal MLIPs, establishing phonon benchmarks as a critical gatekeeper for potential deployment in materials discovery workflows ([arXiv.org](#)).

1.1.2 Distinction between phonon frequencies, density of states, and derived thermodynamic quantities A comprehensive phonon benchmark must distinguish between **three hierarchical levels** of vibrational information, each presenting distinct computational and interpretational challenges. At the most detailed level, **phonon frequencies** ($\omega_{\mathbf{q},\lambda}$) as functions of wavevector $\mathbf{q}$ and branch index λ provide complete dispersion relations, enabling direct comparison with inelastic neutron scattering (INS) and inelastic X-ray scattering (IXS) experiments ([Source](#)). This level captures directional anisotropy, mode crossings, and critical points in the Brillouin zone, but requires substantial computational investment for dense $\mathbf{q}$ -point sampling.

At the intermediate level, the **phonon density of states (PDOS)** $g(\omega)$ integrates over wavevectors to yield a frequency distribution that preserves statistical mode weighting while losing $\mathbf{q}$ -point specificity. The PDOS enables efficient calculation of **thermodynamic properties** through frequency integrals and facilitates comparison with Raman spectroscopy and specific heat measurements ([SourceForge](#)). Critically, PDOS agreement does not guarantee accurate individual frequencies—systematic shifts or mode misassignments may be masked in the integrated representation.

At the coarsest but most application-relevant level, **derived thermodynamic quantities**—including **vibrational entropy** S_{vib} , **Helmholtz free energy** F , and **constant-volume heat capacity** C_V —integrate the PDOS with temperature-dependent Bose-Einstein weighting. These quantities de-

termine finite-temperature phase stability and thermal behavior, yet their accuracy depends on cumulative errors across the entire spectrum. The **PFT (Phonon Fine-Tuning)** study demonstrated that **55% average improvement in phonon frequency errors** propagates to comparable reductions in thermodynamic property errors: Δ _max MAE from 24 K to 10 K, S_{vib} from 32 to 11 J/mol·K, F from 12 to 4 kJ/mol, and C_V from 6 to 2 J/mol·K ([ResearchGate](#)) . This quantitative linkage establishes frequency accuracy as a reliable predictor of integrated property performance.

Property Level	Information Content	Primary Metrics	Application Relevance
Phonon frequencies (q,)	Complete q-resolved dispersion	MAE, RMSE per q-point/ band	Spectroscopy, mode analysis
Phonon density of states g()	Frequency distribution, no q-dependence	Pearson/Spearman correlation, EMD	Thermodynamic integration, Raman
Thermodynamic quantities (S_{vib} , F , C_V)	Temperature-dependent integrals	MAE at specified T, relative errors	Phase stability, thermal management

Table 1: Hierarchical levels of phonon-derived information with corresponding metrics and applications.

1.1.3 Role of phonon benchmarks in assessing interatomic potential transferability and reliability Phonon benchmarks serve a **dual diagnostic function**: they quantify absolute accuracy against reference data and assess **transferability**—the capacity to perform reliably on systems outside training distributions. This second role is particularly critical for universal MLIPs, which claim zero-shot generalization across chemical space yet may exhibit **composition-dependent failure modes** not evident from energy-focused validation ([SourceForge](#)) . The phenomenon of “**force-constant collapse**”—where neural network potentials predict reasonable forces but severely inaccurate curvatures—has been documented across multiple architectures, underscoring the necessity of explicit phonon validation ([SourceForge](#)) .

The **reliability assessment** enabled by phonon benchmarks extends to **dynamical stability prediction**, which is fundamental for materials discovery. A potential’s ability to correctly identify imaginary frequency modes—indicating mechanical instability—directly impacts its utility for structure screening. Large-scale studies reveal significant variation: the **PhononBench** evaluation of **108,843 AI-generated structures** found that only **25.83% were dynamically stable**, with top-performing generative models achieving merely **41.0% stability rates** ([Mendeley Data](#)) . For interatomic potentials, **MatterSim** achieved **~95% true-positive rate** for stability classification, approaching DFT reliability at fractionally reduced cost ([arXiv.org](#)) . These metrics contextualize what constitutes “good” performance and highlight the importance of systematic benchmarking at scale.

1.2 Crystalline Inorganic Materials Focus

1.2.1 Structural diversity: elemental solids, oxides, alloys, and complex ceramics The GLIM project’s focus on **crystalline inorganic materials** encompasses structural diversity that challenges interatomic potentials across bonding types and coordination environments. **Elemental solids** span metallic bonding (fcc Al, bcc Fe, hcp Mg), covalent bonding (diamond Si, graphite C), and van der Waals bonding (layered materials), each with distinct phonon characteristics: metals with nearly free-electron-like acoustic dispersion, covalent materials with high-frequency optical modes reflecting strong directional bonds, and van der Waals materials with extreme frequency separation between stiff intralayer and soft interlayer modes ([nih.gov](#)) .

Oxide materials introduce ionic bonding, polar phonon modes, and **LO-TO splitting** at zone centers requiring non-analytical correction terms. Perovskite oxides like SrTiO₃ exhibit **soft modes** associated with ferroelectric instabilities that are exceptionally sensitive to potential accuracy ([IUCr Crystallography Journals Online](#)) . **Alloys and intermetallic compounds** present compositional disorder and local environment variations, with phonon spectra reflecting both mass disorder effects and chemical ordering. **Complex ceramics**—including zeolites, garnets, MAX phases, and superionic conductors—extend to large unit cells with hundreds of atoms, where low-frequency framework modes and high-frequency cation vibrations may show divergent potential accuracy ([arXiv.org](#)) .

Material Class	Bonding Character	Phonon Challenges	Representative Systems
Elemental metals	Metallic, delocalized	Kohn anomalies, Fermi surface effects	Al, Cu, Fe, Mg
Covalent semiconductors	Directional, strong	High-frequency optical modes, flat TA branches	Si, Ge, diamond
Van der Waals materials	Layered, anisotropic	Extreme frequency range, interlayer softness	Graphite, MoS ₂ , h-BN
Ionic oxides	Polar, charge transfer	LO-TO splitting, dielectric response	MgO, Al ₂ O ₃ , SrTiO ₃
Complex ceramics	Mixed bonding, large cells	Mode localization, computational cost	Zeolites, garnets, MAX phases

Table 2: Structural diversity in crystalline inorganic materials with phonon-specific challenges.

1.2.2 Chemical space coverage relevant to technological applications The **12,000-material scale** of the GLIM project enables systematic coverage of chemical space relevant to **energy, electronics, and structural applications**. Priority domains include: **battery materials** (LiCoO₂, LiFePO₄, Li₁₀La₆Zr₂O₂₁) where phonon properties control ionic conductivity and thermal stability; **thermoelectrics** (PbTe, Bi₂Te₃, half-Heusler compounds) where phonon engineering optimizes the electronic-thermal transport trade-off; **semiconductors and dielectrics** (GaAs, GaN, HfO₂) where phonon scattering limits carrier mobility; and **high-temperature structural materials** (Ni-based superalloys, refractory nitrides) where vibrational contributions to free energy determine phase stability ([alphaxiv.org](#)) .

Large-scale benchmarks have revealed **systematic composition-dependent performance variations**. The **MatterSim benchmark** of >10,000 materials found that **transition metal compounds generally exhibit larger phonon errors than main-group materials** due to complex d-electron bonding, while **heavy elements and rare-earth containing systems remain challenging** due to limited training data ([arXiv.org](#)) . The **PhononBench** analysis showed **oxygen-dominated elemental distribution** (10,272 occurrences) with noble gases essentially absent, highlighting gaps in coverage of technologically important elements ([arXiv.org](#)) . For GLIM, intentional inclusion of underrepresented species—heavy transition metals, actinides, lanthanides—will test potential performance in data-scarce regimes.

1.2.3 Challenges in phonon calculations for materials with varying bonding character Different bonding characters present **distinct methodological challenges** for phonon benchmarking. **Metallic systems** require dense k-point sampling for electronic convergence and proper treatment of **Kohn anomalies** from Fermi surface nesting; potentials lacking explicit electronic structure may capture these

effects only implicitly through training data. **Covalent materials** with directional bonds demand precise description of **bond-angle dependencies**; the flat transverse acoustic branches in diamond-structure semiconductors are particularly sensitive to angular force terms (Vincent Legat) .

Van der Waals materials present perhaps the **most severe challenge**: interlayer shear and breathing modes at **sub-10 cm⁻¹** frequencies require accurate long-range dispersion treatment, while intralayer modes exceed **1500 cm⁻¹**, demanding numerical precision across **two orders of magnitude** (ChemRxiv) . Most uMLIPs trained on PBE data without explicit dispersion corrections exhibit **substantially larger errors for van der Waals systems**—the exception being **ALIGNN-FF**, explicitly trained on JARVIS-DFT data with vdW-DF-optB88 functional (IOPscience) . **Ionic and polar materials** require **Born effective charges and dielectric tensors** for LO-TO splitting; potentials lacking explicit charge descriptions must learn effective screening implicitly, with variable success.

1.3 The GLIM Project Context

1.3.1 Scale: 23 interatomic potentials across 12,000 materials The GLIM project’s **23 potentials × 12,000 materials = 276,000 phonon calculations** represents an **unprecedented systematic comparison**. This scale exceeds prior benchmarks: the **MatterSim benchmark** evaluated **7 uMLIPs on >10,000 materials** (arXiv.org) ; the **comprehensive uMLIP study** assessed **12 potentials on ~5,000 crystals** (AIP Publishing) ; and **PhononBench** evaluated stability for **108,843 structures with single-potential (MatterSim-v1) screening** (chrismarianetti.org) . The expanded potential count enables **architectural comparison** across: graph neural networks (CHGNet, M3GNet), equivariant neural networks (MACE, NequIP, SevenNet), transformer-based models (eqV2, ORB), and classical empirical potentials (EAM, MEAM, ReaxFF).

Potential Family	Representative Models	Key Architectural Features	Typical Phonon _max MAE
Equivariant GNNs	MACE-MP-0, MACE-MPA-0, SevenNet-0	E(3)-equivariant message passing, many-body correlations	2–6 meV (IOPscience)
Invariant GNNs	CHGNet, M3GNet, MatGL	Crystal graph convolutions, attention mechanisms	6–15 meV (arXiv.org)
Transformer-based	ORB, eqV2-M, OMat24	Self-attention, global receptive field	2–8 meV (Source)
Specialized/fine-tuned	PFT-Nequix, MatterSim-FT	Phonon-targeted training, active learning	<2 meV (arXiv.org)
Classical empirical	EAM, MEAM, ReaxFF	Fixed functional forms, element-specific fitting	20–100+ meV (chrismarianetti.org)

Table 3: Interatomic potential families with architectural features and typical phonon accuracy ranges.

1.3.2 Objectives: systematic accuracy assessment and computational efficiency evaluation The **dual objectives** of accuracy and efficiency assessment reflect practical constraints in materials modeling. **Accuracy metrics** must span hierarchical levels: **primary metrics** (frequency MAE, _max error, stability prediction accuracy) for global ranking; **secondary metrics** (PDOS correlation, thermodynamic property errors) for detailed pattern analysis; and **diagnostic metrics** (composition-dependent errors,

failure mode categorization) for improvement guidance. **Efficiency evaluation** encompasses single-point inference cost, parallel scalability, and workflow integration overhead—factors that determine practical deployability.

The **accuracy-efficiency trade-off** is not monotonic: the **CHIPS-FF benchmark** found that **ORB** achieves accuracy comparable to **OMat24** with close to an order of magnitude more computational efficiency ([NIST](#)) , while **OMat** exhibits drastically steeper scaling with system size than **MACE** or **SevenNet** despite similar small-system accuracy ([RSC Publishing](#)) . Systematic quantification of these trade-offs enables **Pareto-optimal potential selection** for specific application constraints.

1.3.3 Integration with existing benchmark ecosystems (JARVIS-Leaderboard, Materials Project) Strategic ecosystem integration amplifies GLIM’s impact and ensures methodological continuity. **JARVIS-Leaderboard** provides standardized evaluation frameworks with automated scoring and public comparison; **CHIPS-FF** demonstrates seamless integration with open-source release of benchmarking tools ([Nature](#)) . **Materials Project** offers PBE-based reference data and extensive community infrastructure, though **functional alignment** with JARVIS-DFT’s **vdW-DF-optB88** requires careful handling. The **MatBench Discovery** framework connects phonon-derived stability metrics with experimental synthesizability predictions, validating practical relevance ([lanl.gov](#)) .

2. Phonopy Workflows for High-Throughput Phonon Calculations

2.1 Core Methodology: Finite Displacement Approach

2.1.1 Supercell generation and symmetry analysis The **finite displacement method** in **Phonopy** ([NIST](#)) constructs force constant matrices by systematic atomic displacement and force calculation. **Supercell dimensions** must exceed the **cutoff radius of interatomic interactions**, with **12 Å minimum dimension** standard for converged long-range forces ([nih.gov](#)) . **Symmetry analysis via spglib** reduces independent displacements by identifying equivalent atoms, achieving **2–10× reduction** depending on crystal symmetry. For high-throughput execution across 12,000 materials, **automated supercell generation with fallback handling** for low-symmetry or large-cell systems is essential.

2.1.2 Atomic displacement magnitudes: standard 0.01–0.03 Å range and convergence considerations **Displacement magnitude selection** balances numerical stability against harmonic approximation validity. **Standard range: 0.01–0.03 Å**, with **0.01 Å typical for DFT** (low force noise) and **0.02–0.03 Å for empirical potentials** (larger signal-to-noise) ([Nature](#)) . Critical **architecture-dependent behavior** was identified by **CHIPS-FF**: **MACE, MatterSim, SevenNet show consistent phonon MAE across displacements**, while **ORB and OMat exhibit drastic MAE increase at small displacements**—attributed to **direct force output versus energy-gradient backpropagation** ([arXiv.org](#)) . This finding mandates **displacement convergence testing per potential family** for valid comparison.

Displacement (Å)	Typical Use Case	Advantages	Risks
0.01	DFT reference, high-precision MLIP	Minimal anharmonic contamination	Force noise for noisy potentials
0.02	Balanced MLIP evaluation	Good signal-to-noise trade-off	Moderate anharmonicity

Table 4 – continued

Displacement (Å)	Typical Use Case	Advantages	Risks
0.03	Empirical potentials, screening	Robust force signals	Significant anharmonic errors

Table 4: Displacement magnitude trade-offs in finite-difference phonon calculations.

2.1.3 Force calculation via interatomic potentials versus density functional theory The **fundamental acceleration** in phonon benchmarking comes from replacing **DFT force evaluation** (minutes to hours per configuration, $O(N^3-N)$ scaling) with **interatomic potential inference** (milliseconds to seconds, **near-linear scaling**). **DFT reference calculations** for 220 materials with anharmonic force constants required **~480,000 CPU-hours** ([arXiv.org](#)) ; **ML-assisted reduction to ~12,000 CPU-hours** ($40\times$ speedup) preserves accuracy within 10% for thermal conductivity ([ACS Publications](#)) . **Fully pre-trained uMLIPs achieve 10^3 – $10\times$ single-point speedup**, enabling thousand-fold scale expansion.

2.2 Geometry Optimization Protocols

2.2.1 Internal coordinate relaxation with fixed lattice parameters Fixed-lattice relaxation decouples phonon accuracy assessment from volume prediction capability. The **comprehensive uMLIP benchmark** explicitly adopted this protocol: “**we did not relax the volume or shape of the unit cells...because our goal was to benchmark and compare the phonon properties calculated on the same crystal structures**” ([Ceder Group](#)) . This ensures that frequency differences reflect **force constant accuracy** rather than **Grüneisen-parameter-shifted volumes**. The **FIRE algorithm** with **0.005 eV/Å force convergence** provides efficient, robust relaxation ([arXiv.org](#)) .

2.2.2 FIRE algorithm implementation with force convergence criteria (0.005 eV/Å typical) **FIRE (Fast Inertial Relaxation Engine)** combines molecular dynamics propagation with adaptive velocity mixing for rapid convergence ([ACS Publications](#)) . **Typical criteria: 0.005 eV/Å maximum force component**, balancing cost against precision. Tighter convergence (0.001 eV/Å) may be needed for materials with **flat PES regions or soft modes**. **ASE integration** enables standardized interfaces across potential backends, with **FrechetCellFilter preferred over ExpCellFilter for lattice parameter accuracy** ([Source](#)) .

2.2.3 Decoupling of structural relaxation from phonon property benchmarking Explicit **decoupling enables error decomposition**: phonon accuracy at **DFT-equilibrium structures** tests pure force constant prediction, while **self-relaxed structure evaluation** tests combined structure+phonon performance. The **PFT study** demonstrated that **fine-tuning on phonon data improves curvature accuracy with minimal degradation of other properties** ([arXiv.org](#)) , validating targeted optimization strategies. For GLIM, **two-tier evaluation** (fixed-structure primary, self-relaxed secondary) provides comprehensive assessment.

2.3 Phonon Spectrum Computation Pipeline

2.3.1 Force constant matrix construction and symmetrization Force constant construction via finite differences: $\Phi_{ij} = -[F_i(+j) - F_i(-j)]/(2)$. **Symmetrization** enforces:

acoustic sum rules (translational invariance), **crystal symmetry constraints**, and **rotational invariance** (for non-periodic directions). Quality metrics include **acoustic mode frequencies at Γ** (should be <10 THz) and **sum rule violation magnitude**. For MLIPs, **automatic differentiation enables analytical Hessians** with exact constraint satisfaction, though at $O(N^2)$ memory cost ([Nature](#)) .

2.3.2 High-symmetry q-point path generation via Seekpath or pymatgen Standardized path generation via **Seekpath** or **pymatgen** ensures reproducible dispersion visualization ([cornell.edu](#)) . **Path density: ~ 100 points per high-symmetry segment** for visualization quality; **band-resolved MAE analysis** identifies mode-specific accuracy patterns. **Automatic generation** handles conventional/primitive cell conventions, with **manual override capability** for non-standard cases.

2.3.3 Fourier interpolation for dense q-point sampling (1000 points/ $\text{\AA}^3$ density) Fourier interpolation extends discrete force constants to arbitrary q-points. **Standard density: 1000 points/ $\text{\AA}^3$** ensures converged PDOS and thermodynamic properties ([arXiv.org](#)) . **Convergence verification:** explicit supercell calculations at commensurate q-points should match interpolated values within tolerance. **Adaptive refinement** benefits materials with sharp spectral features or flat bands.

2.3.4 Phonon density of states calculation via uniform q-point meshes PDOS integration via **uniform meshes** ($20 \times 20 \times 20$ to $40 \times 40 \times 40$ typical) with **tetrahedron method** or **Gaussian smearing** (~ 0.1 – 0.5 THz). **Tetrahedron methods** preserve van Hove singularities; **Gaussian smearing** provides robustness for coarse meshes. The PDOS enables **direct comparison with vibrational spectroscopies** and **thermodynamic property integration**.

2.4 Dynamical Stability Assessment

2.4.1 Identification of imaginary frequency modes Dynamical stability requires $\omega^2 > 0$ for all q . **Numerical threshold:** $|\omega| < 0.1$ THz (~ 0.4 meV) typically considered noise; **clearly negative values indicate true instability**. **Spatial distribution matters:** **Γ -point instabilities** suggest ferroelectric/structural transitions; **zone-boundary instabilities** indicate competing periodic structures. **Automated flagging** with manual review for borderline cases ensures reliable classification.

2.4.2 Stability rate metrics for large-scale screening **Stability rate**—fraction correctly classified as stable/unstable—enables rapid potential comparison. **PhononBench baseline: 25.83% stable** across 108,843 AI-generated structures; **top generative model (MatterGen): 41.0%** ([nist.gov](#)) . For interatomic potentials, **MatterSim: $\sim 95\%$ true-positive rate** for DFT-stable materials ([Source](#)) . **Confusion matrix decomposition** (true/false positives/negatives) guides application-specific selection: **false positives waste experimental effort; false negatives miss viable candidates**.

2.4.3 Handling of soft modes and structural phase transitions **Soft modes** ($\omega \rightarrow 0$) indicate **proximity to phase transitions** and challenge numerical precision. **Reduced displacement magnitudes** may be needed for harmonic validity; **anharmonic treatments** (self-consistent phonons, molecular dynamics) may be required for finite-temperature stability. **Benchmark documentation** should flag strongly anharmonic materials and specify harmonic approximation limitations.

3. Reference Data: JARVIS-DFT Phonon Database

3.1 Database Characteristics and Coverage

3.1.1 vdW-DF-optB88 functional basis for phonon calculations JARVIS-DFT employs vdW-DF-optB88, distinguishing it from PBE-dominant Materials Project ([arXiv.org](https://arxiv.org)) . The optB88 exchange optimization improves van der Waals binding while maintaining reasonable performance for covalent/ionic materials. **Functional consequences:** 1–2% lattice parameter differences versus PBE for dense solids, substantially larger for layered materials; corresponding phonon frequency shifts of 2–6% (dense) to 20%+ (van der Waals) ([NIST](https://nist.gov)) . This functional specificity must be acknowledged when comparing PBE-trained potentials to optB88 references.

3.1.2 Material diversity and chemical space representation JARVIS-DFT encompasses ~90,000 materials with 17,402 having elastic tensors and phonons (nist.gov) . Coverage spans: bulk crystals (metals, semiconductors, insulators); 2D materials (graphene, TMDs, boron nitride); topological materials; and catalysts. **Chemical diversity:** 89 elements with strength in energy-relevant materials (battery electrodes, photovoltaics, thermoelectrics). For GLIM’s 12,000 materials, stratified sampling ensures representative coverage across crystal systems, bonding types, and application domains.

3.1.3 Validation against experimental phonon data where available Experimental validation—inelastic neutron scattering, Raman, infrared—provides ground truth assessment. Typical DFT accuracy: 5–10% for well-characterized systems, with larger deviations for strongly anharmonic or correlated electron systems ([arXiv.org](https://arxiv.org)) . INS studies (e.g., Cu O single-crystal) represent gold-standard validation ([arXiv.org](https://arxiv.org)) . JARVIS-DFT systematic validation enables uncertainty quantification for benchmark interpretation.

3.2 Integration with Benchmarking Workflows

3.2.1 Automated structure matching and comparison protocols Structure matching via pymatgen StructureMatcher identifies equivalent structures despite cell choice variations, origin shifts, or small numerical deviations. **Tolerance parameters:** typically 0.2 Å and 5° for lattice/angle matching. **Match success rates:** ~80% for curated test sets (e.g., 84/104 in CHIPS-FF) ([RSC Publishing](https://rsc.org)) ; failed matches indicate genuine structural differences (polymorphs, convergence to distinct minima) requiring manual review or exclusion.

3.2.2 Consistency checks for functional-dependent phonon properties PBE-optB88 comparison quantifies functional sensitivity. Systematic characterization across representative subsets enables: functional-corrected metrics; material-class-specific uncertainty; and potential performance normalization. Cross-functional validation (materials with both PBE and optB88 calculations) provides direct functional difference estimation.

3.2.3 Handling of discrepancies between PBE and optB88 reference data Discrepancy magnitude and distribution guide reference data weighting. Large discrepancies (>10% frequency shifts) for layered/molecular materials indicate functional uncertainty dominating potential error; small discrepancies for dense covalent solids enable cleaner accuracy assessment. **GLIM strategy:** primary benchmark against optB88 (JARVIS-DFT); PBE subset recalculation for direct PBE-trained potential evaluation; explicit functional difference documentation.

3.3 Alternative Reference Datasets

3.3.1 Materials Project phonon data (PBE functional) Materials Project provides PBE phonon data with extensive community validation and MPContribs integration ([Source](#)) . Coverage: thousands of materials with automated workflow consistency. Direct alignment with PBE-trained uMLIP training data reduces functional mismatch errors. Complementary to JARVIS-DFT: overlapping materials enable cross-validation; unique materials extend benchmark coverage.

3.3.2 MDR (Materials Data Repository) phonon benchmark MDR phonon benchmark: ~10,000 materials with PBE phonons, explicit quality control, reproducibility documentation ([arXiv.org](#)) . Used for PFT validation with held-out test sets. Fine-tuning demonstration: 55% average phonon property error reduction ([ACS Publications](#)) . Curated design makes MDR valuable for targeted accuracy assessment, though smaller scale than JARVIS-DFT.

3.3.3 PhononDB and other community resources PhononDB and specialized databases (AFLOWLIB, OQMD) provide additional coverage with varying quality control. Experimental phonon compilations (from INS facilities) enable ultimate validation. Community-driven benchmark evolution, exemplified by PhononBench open release of 108,843 structure calculations ([Source](#)) , enables cumulative improvement and collaborative validation.

4. Error Metrics for Phonon Frequency Spectrum Comparison

4.1 Direct Frequency Comparison Metrics

4.1.1 Mean absolute error (MAE) per q-point or band-resolved $MAE_{q,b} = \frac{\sum_i |f_i - \hat{f}_i|}{n}$ with q-point weights w_j accounting for non-uniform sampling ([Source](#)) . Band-resolved decomposition: acoustic mode MAE tests long-wavelength elastic response; optical mode MAE tests short-range bonding description. Typical uMLIP performance: 2–10 meV for well-represented materials, 10–50 meV for challenging classes (transition metals, heavy elements) ([Source](#)) .

Potential	max MAE (meV)	Mean MAE (meV)	Max MAE (meV)
MACE-MP-0	2.7	3.71	17.22 (Source)
CHGNet	6.2	8.12	37.34 (arXiv.org)
M3GNet	7.8	10.41	52.18 (Source)
Nequix MP (PFT)	1.0	2.2	5.8 (Source)
AIREBO (classical)	29.5	36.7	89.2 (Source)

Table 5: Representative phonon MAE values from large-scale benchmarks.

4.1.2 Root mean square error (RMSE) for frequency distributions $RMSE = \sqrt{\sum_i (f_i - \hat{f}_i)^2 / n}$ emphasizes large errors and outliers. RMSE/MAE ratio indicates error distribution shape: ~1 for Gaussian errors, »1 for heavy-tailed distributions with catastrophic failures. Critical for identifying potential-specific failure modes that may dominate thermodynamic properties despite modest typical errors.

4.1.3 Maximum frequency error ($\omega_{\max}$ MAE) as a scalar diagnostic $\omega_{\max}$ MAE = $|\max(\omega) - \max(\hat{\omega})|$ provides **single-number accuracy assessment**. **Physical relevance:** **bond stiffness descriptor**, **Debye temperature correlate**, **experimentally accessible via Raman**. **PFT improvement:** **24 K $\rightarrow$ 10 K** (2.1 $\rightarrow$ 0.9 meV) demonstrates **targeted training efficacy** (Source) . **Computational efficiency:** requires only **Γ -point or limited q-point sampling** for rapid screening.

4.1.4 Frequency-dependent weighting schemes **Application-specific weighting** enables **targeted potential selection**:

Weighting Function	Application	Frequency Emphasis
$w(\omega) = 1$ (uniform)	General benchmark	All frequencies equally
$w(\omega) = 1/\omega$	Low-frequency emphasis	Thermal transport, acoustic modes
$w(\omega) = n_D(\omega, T)$	Temperature-dependent	Thermodynamic properties at T
$w(\omega) = \exp(-\beta \hbar \omega)$	Spectroscopic matching	Specific experimental probe

Table 6: Frequency-dependent weighting schemes for application-specific benchmarking.

4.2 Spectral Similarity Measures

4.2.1 Pearson correlation coefficient for phonon density of states $r = \text{Cov}(g, \hat{g}) / (\sigma_g \sigma_{\hat{g}})$ quantifies **linear PDOS shape agreement**. **Values >0.95 :** excellent shape reproduction; **0.8–0.95:** good with systematic shifts; **<0.8 :** poor spectral fidelity. **Insensitive to uniform frequency shifts**—complements pointwise MAE. **uMLIP typical:** $r > 0.95$ for well-performing models (Source) .

4.2.2 Spearman rank correlation for mode ordering preservation $\rho_s = 1 - 6 \sum d_i^2 / [n(n^2-1)]$ assesses **mode ordering preservation** regardless of absolute scale. **High ρ_s :** correct relative stiffness of different modes; **low ρ_s :** mode crossings or misassignments. Critical for **spectroscopic applications** where mode character determines selection rules. **INS benchmarking** employed Spearman for **PDOS similarity quantification** (Source) .

4.2.3 Earth mover’s distance and other distribution comparison metrics **Earth mover’s distance (Wasserstein metric):** intuitive frequency units, sensitive to **peak position shifts and intensity redistribution**. **Alternative:** **Kullback-Leibler divergence** (information-theoretic), **Jensen-Shannon divergence** (symmetrized, bounded). **Computational cost** may limit large-scale application; **sampling or approximation** enables practical deployment.

4.3 Thermodynamic Property Errors

4.3.1 Vibrational entropy (S_{vib}) MAE and relative errors $S_{\text{vib}}(T) = k_B \sum_i g_i [(n_i + 1) \ln(n_i + 1) - n_i \ln n_i]$ d. Typical errors: **0.1–0.5 k_B /atom** for uMLIPs, **>1 k_B /atom** for classical potentials (Source) . **Temperature dependence:** **low-T dominated by acoustic modes**, **high-T by full spectrum**. **PFT improvement:** **32 $\rightarrow$ 11 J/mol·K** (66% reduction) (Source) .

4.3.2 Helmholtz free energy (F) accuracy assessment $F_{\text{vib}}(T) = k_B T \sum_i g_i \ln[2 \sinh(\hbar \omega_i / 2k_B T)]$ d determines **finite-temperature phase stability**. **Error propagation:** **~ 0.1 eV/atom**

uncertainty from 0.1 k_B/atom S_{vib} error at 1000 K—comparable to DFT formation energy precision. Relative phase energy errors more critical than absolute for structure prediction.

4.3.3 Heat capacity at constant volume (C_v) error quantification $C_V(T) = k_B \sum_i g_i \left(\frac{\epsilon_i}{k_B T} \right)^2 e^{\epsilon_i / k_B T} / (e^{\epsilon_i / k_B T} - 1)^2$. Low-T C_V ∝ T³: acoustic mode accuracy test; high-T → 3Nk_B: classical limit verification. PFT improvement: 6 → 2 J/mol·K MAE ([Semantic Scholar](#)). Deviations from expected T-dependence indicate frequency spectrum errors.

4.3.4 Temperature-dependent error propagation Multi-temperature evaluation (100 K, 300 K, 600 K, 1000 K standard) reveals which frequency ranges dominate inaccuracies. Non-monotonic error variation indicates mode-specific compensation or amplification. Thermal conductivity study found accurate harmonic frequencies do not guarantee correct temperature dependence, highlighting anharmonicity as additional challenge ([Semantic Scholar](#)).

4.4 Composite and Rank-Based Metrics

4.4.1 greedyRank multi-objective ranking for potential comparison greedyRank algorithm: iteratively select non-dominated potentials, remove from consideration, repeat for tiered ranking ([ResearchGate](#)). Respects Pareto optimality: no arbitrary metric weighting. Enables application-specific selection based on accuracy-cost trade-offs relevant to user's constraints.

4.4.2 Success rates for correct symmetry mode identification Symmetry mode identification: acoustic vs. optical, transverse vs. longitudinal, IR-active vs. Raman-active. Automated analysis via phonopy mode decomposition with group theory tools. Success rate metrics assess eigenvector accuracy beyond frequencies—critical for spectroscopic intensity predictions.

4.4.3 Stability prediction accuracy metrics Binary classification metrics: precision, recall, F1, ROC-AUC. Class imbalance (baseline ~25% stable) requires careful interpretation: random guessing achieves 25% accuracy; perfect prediction achieves 100%. Skill scores (improvement over baseline) enable fair comparison. MatterSim: ~95% true-positive rate with validated false-positive/negative characterization ([arXiv.org](#)).

5. Potential Family Performance Comparison

5.1 Universal Machine Learning Interatomic Potentials (uMLIPs)

5.1.1 MatterSim: DFT-level accuracy demonstrated across >10,000 materials MatterSim represents leading uMLIP performance with explicit phonon-focused validation. Scale: >10,000 materials with DFT-level accuracy claims ([AIP Publishing](#)). Benchmark findings: intermediate performance in comprehensive thermal conductivity study (MAE 0.347 log(W/m·K)), with noted dissociation between force accuracy and IFC prediction accuracy—“impressive IFC prediction despite worst force accuracy” ([MDPI](#)). PhononBench stability screening: ~95% true-positive rate ([ResearchGate](#)). Architecture: non-equivariant graph neural network with active learning across temperature (0–5000 K) and pressure (0–1000 GPa) ([ResearchGate](#)).

5.1.2 MACE-MP-0: Message-passing neural network performance MACE-MP-0 achieves top-tier phonon accuracy through E(3)-equivariant message passing with explicit many-body correlations. Systematic benchmark: minimum MAE 0.31 meV, mean 3.71 meV, maximum 17.22 meV—best among tested uMLIPs (ResearchGate) . Consistent performance across displacement magnitudes indicates robust force prediction in linear response regime (ResearchGate) . Newer MACE-MPA-0 shows improved phonon results, demonstrating rapid architectural evolution (Semantic Scholar) .

5.1.3 CHGNet: Crystal graph convolutional network benchmarks CHGNet exhibits solid but not exceptional phonon performance: mean MAE 8.12 meV, maximum 37.34 meV (Directory of Open Access Journals) . Thermal conductivity study: “worst model...in terms of predicting accuracy of the LTC” with MAE 0.905 log(W/m·K) and systematic underprediction (ResearchGate) . Smaller parameter count (~400k) enables rapid inference but may limit representational capacity for complex phonon properties. Architecture: graph attention with explicit three-body interactions and coordinate incorporation.

5.1.4 SevenNet-0, ORB, eqV2-M, and emerging architectures

Model	Key Features	Phonon Performance	Efficiency Notes
SevenNet-0 / SevenNet-MP-ompa ORB v3	Equivariant, parallelized message passing Smooth overlap atomic positions, direct force output	Top-tier , near-MACE accuracy (Nature) Highest ranking in uMLIP benchmark (chrismarianetti.org)	Favorable scaling ~10× more efficient than OMat24 (arXiv.org)
ORB v1 eqV2-M / EquiformerV2	Earlier version Equivariant transformers, higher-order representations	Good, below v3 Strong , improved with fine-tuning (FT: MAE 0.174 log(W/m·K)) (arXiv.org)	Similar efficiency Architecture-dependent overhead
OMat24 / GRACE-2L-OAM	Large-scale training, diverse data	Top-tier accuracy (Nature)	Drastically steeper scaling than MACE/SevenNet (Nature)

Table 7: Emerging uMLIP architectures with phonon performance and efficiency characteristics.

Critical observation: ORB and OMat show displacement-dependent phonon errors—drastic MAE increase at small displacements attributed to direct force output versus energy-gradient backpropagation (nih.gov) . This architectural choice trade-off affects numerical stability in harmonic regime.

5.1.5 Comparative accuracy: typical MAE ranges of 2–10 meV for __max

Accuracy Tier	$_\text{max}$ MAE Range	Representative Models	Application Suitability
Exceptional	<2 meV	PFT-fine-tuned, specialized	Precision thermodynamics, spectroscopy
Excellent	2–5 meV	MACE-MP-0, ORB v3, SevenNet-0, MatterSim	General-purpose, high-throughput screening
Good	5–10 meV	CHGNet, M3GNet, base eqV2	Screening, qualitative trends
Moderate	10–20 meV	Older/incomplete training	Rough estimates, not recommended for precision
Poor	>20 meV	Classical potentials (untrained), failed models	Not suitable for quantitative predictions

Table 8: Phonon accuracy tiers with representative models and application guidance.

5.2 Fine-Tuned and Specialized ML Potentials

5.2.1 Phonon fine-tuning (PFT) approaches for improved accuracy Phonon Fine-Tuning (PFT) demonstrates **dramatic accuracy gains through targeted training on phonon data**. **Methodology:** direct Hessian supervision against DFT force constants with **stochastic column sampling** for scalability ([arXiv.org](#)) . **Training cost:** ~140 A100-hours with co-training (prevents catastrophic forgetting), ~60 A100-hours without—“small fraction of other competitive MPtrj-trained MLIPs” ([Source](#)) . **Key innovation:** Hessian-vector products reduce complexity from $O(N^2)$ to $O(N)$ per step.

5.2.2 Domain-specific training for thermal transport applications Beyond generic phonon accuracy, thermal transport requires anharmonic properties—third-order force constants for phonon-phonon scattering. **PFT improvement in Matbench Discovery $_\text{SRME}$: 0.446 $\rightarrow$ 0.306** (31% improvement), **state-of-the-art among MPtrj-trained models** ([arXiv.org](#)) . This transfer from harmonic Hessian training to anharmonic properties suggests **curvature supervision captures essential physics extending beyond harmonic regime**.

5.2.3 Accuracy gains: sub-meV errors achievable with targeted optimization Fine-tuned achievements: $_\text{max}$ MAE 10 K (~0.9 meV), S_vib MAE 11 J/mol·K, F MAE 4 kJ/mol, C_V MAE 2 J/mol·K ([Source](#)) . Approaching practical limits set by reference data quality and numerical precision. Diminishing returns for further improvement suggest **near-convergence for harmonic properties; anharmonic and finite-temperature effects become next frontier**.

5.3 Classical Empirical Potentials

5.3.1 Embedded atom method (EAM) and modified EAM (MEAM) performance EAM/MEAM potentials achieve **computational efficiency through simple functional forms** but suffer limited transferability. Typical phonon errors: 20–100+ meV for $_\text{max}$ in untrained

systems, 5–20 meV for well-parameterized elemental systems ([Source](#)) . Silicon carbide benchmark: EAM/MEAM ϵ_{max} errors 200–800 cm⁻¹ (25–100 meV), Debye temperature errors 50–200 K ([arXiv.org](#)) . Strength: near-linear scaling, analytical forces, decades of parameterization experience.

5.3.2 Angular-dependent potentials (ADP) for transition metals ADP extensions improve transition metal description through explicit angular terms. Moderate phonon accuracy improvements over EAM, but systematic phonon optimization rare—typically fitted to equilibrium properties and elastic constants rather than explicit frequency data. Parameterization effort substantial; accuracy gains often not commensurate with complexity increase.

5.3.3 Reactive force fields (ReaxFF) and bond-order potentials ReaxFF enables chemistry-dependent bonding for reactive dynamics but computational cost approaches neural networks with inferior phonon accuracy. Fe-MLIP comparison: substantial vibrational frequency discrepancies versus ML potentials ([Source](#)) . Trade-off: reactivity versus harmonic accuracy—ReaxFF prioritizes bond breaking/forming over precise equilibrium curvature.

5.3.4 Typical error magnitudes: 20–100+ meV for ϵ_{max} in untrained systems Classical potential phonon errors—one to two orders of magnitude larger than best ML potentials—reflect fundamental representational limitations. Within fitting domains, errors can be minimized through extensive parameterization, but general transferability remains poor. Continued relevance: specialized applications where training data unavailable, extreme conditions beyond ML coverage, or educational/interpretive purposes.

5.4 Cross-Family Comparative Analysis

5.4.1 Systematic accuracy hierarchies: MLIPs > classical potentials for general systems Clear hierarchy emerges: universal MLIPs achieve 2–10 meV typical accuracy, fine-tuned variants <2 meV, classical potentials 20–100+ meV for general systems. Gap largest for systems far from classical fitting domains. Computational cost hierarchy inverts: classical potentials fastest, efficient MLIPs (ORB, MACE) intermediate, large architectures (OMat) slowest. Pareto frontier identifies optimal trade-offs for specific constraints.

5.4.2 Composition-dependent performance variations Systematic patterns: main-group compounds < transition metals < heavy elements/rare earths in typical accuracy. Van der Waals materials: large errors for PBE-trained potentials versus optB88 references; ALIGNN-FF (JARVIS-trained) exceptional. Oxides with strong correlation: challenging for all potentials; explicit charge/magnetic treatment may help. Identifying these patterns enables application-specific potential selection.

5.4.3 Transferability limitations and failure mode analysis Common failure modes: spurious imaginary frequencies (overly soft potentials), missing soft modes (overly stiff), incorrect mode ordering (symmetry errors), catastrophic errors for specific chemistries (training gaps). CHIPS-FF tracking: convergence success rates, materials with specific potential failures ([Source](#)) . Diagnostic value: guides targeted improvement and safe application domains.

6. Computational Cost and Scalability Analysis

6.1 Cost Relative to Density Functional Theory

6.1.1 DFT phonon calculation costs: ~480,000 CPU-hours for 220 materials (anharmonic) Direct DFT phonon calculations impose substantial computational burden: ~480,000 CPU-hours for 220 materials with anharmonic force constants (eScholarship) . Harmonic calculations somewhat less expensive but still prohibitive for 12,000-material scale. Cost drivers: multiple self-consistent field calculations per material, supercell size requirements, dense k-point sampling for metals.

6.1.2 ML-accelerated reductions: ~12,000 CPU-hours for equivalent scale ML-assisted approaches achieve dramatic reduction: ~12,000 CPU-hours for equivalent 220-material scale (40× speedup) with thermal conductivity accuracy within 10% (arXiv.org) . Strategy: local ML models trained on sparse DFT predict forces for full displacement set. Trade-off: still requires substantial DFT investment for training data.

6.1.3 Speedup factors: 10–100× for harmonic phonon properties Fully pre-trained uMLIPs enable 10³–10 × single-point speedup versus DFT, 10–100× full workflow speedup accounting for phonon calculation overhead. Absolute time: seconds to minutes per material for complete harmonic phonon analysis with efficient implementations. Critical enabler: GLIM-scale benchmarking feasible only with ML acceleration.

6.2 Interatomic Potential Evaluation Costs

6.2.1 Classical potential computational efficiency: near-linear scaling with system size Classical potentials maintain near-linear O(N) scaling with minimal overhead. Millions of atoms feasible for molecular dynamics; phonon calculations limited by supercell size for force constant convergence rather than per-evaluation cost. Primary advantage: absolute speed and deterministic execution.

6.2.2 Neural network potential inference costs: architecture-dependent overhead Neural network inference introduces architecture-dependent overhead:

Architecture Class	Typical System Size	Time per Evaluation	Scaling
Local descriptors + feedforward	<1000 atoms	~10–100 ms	O(N)
Message-passing GNNs (few layers)	<5000 atoms	~100 ms–1 s	O(N)–O(N log N)
Deep message-passing (many layers)	<2000 atoms	~1–10 s	O(N ²)–O(N ³)
Transformers, global attention	<1000 atoms	~10 s–1 min	O(N ²)–O(N ³)

Table 9: Neural network potential inference characteristics by architecture class.

CHIPS-FF finding: OMat scales drastically more steeply than MACE or SevenNet despite similar small-system accuracy (AIP Publishing) —message-passing depth and global receptive field drive cost-accuracy trade-off.

6.2.3 Message-passing versus local descriptor-based model efficiency Message-passing networks (MACE, SevenNet, CHGNet) achieve superior accuracy through explicit neighbor communication but incur overhead from iterative message passing. Local descriptor models (earlier BPNN, some implementations) sacrifice some accuracy for speed. Optimal choice depends on: required precision, system size, available hardware (GPU acceleration benefits message-passing).

6.3 High-Throughput Workflow Optimization

6.3.1 Parallelization strategies for 12,000-material benchmarks Embarrassingly parallel across: materials, displacements, q-points. Hierarchical parallelization: material-level distribution across nodes, displacement/q-point threading within node. GPU acceleration for neural network inference; CPU clusters for classical potentials and post-processing. Estimated GLIM cost: ~10–10 GPU-hours for complete 23×12,000 benchmark with efficient implementation.

6.3.2 Automated convergence testing and adaptive q-point sampling Adaptive protocols optimize accuracy-cost trade-off: coarse initial sampling, refinement where variance high, early termination for converged properties. Convergence criteria: frequency change <0.1 THz with sampling doubling, PDOS integral change <1%. Particularly valuable for materials with complex spectra where uniform dense sampling wasteful.

6.3.3 Resource allocation for 23-potential comparative studies Stratified execution: all potentials on representative subset (e.g., 1000 materials) for statistical power; top performers on full set for definitive ranking; specialized analysis (fine-tuning, failure modes) on priority subsets. Incremental release: preliminary results inform community feedback and protocol refinement.

6.4 Cost-Accuracy Trade-offs

6.4.1 Optimal potential selection for screening versus precision applications

Application Type	Priority	Recommended Potential Class	Rationale
Large-scale screening (10 + materials)	Speed, robustness	Efficient uMLIP (ORB, MACE-MP-0)	~10× faster than best accuracy, good enough
Targeted discovery (10 ³ –10 materials)	Balanced accuracy-cost	Top-tier uMLIP (SevenNet-0, MACE-MPA-0)	Near-best accuracy, acceptable cost
Precision thermodynamics (10 ² –10 ³ materials)	Maximum accuracy	Fine-tuned specialist (PFT variant)	Sub-meV errors where needed
Extreme conditions, novel chemistry	Robustness, extrapolation	Ensemble, active learning, or DFT fallback	Uncertainty quantification critical

Table 10: Application-specific potential selection guidance.

6.4.2 Hierarchical benchmarking: coarse-to-fine potential evaluation Three-tier strategy: **Tier 1**—fast metrics (ϵ_{max} , stability rate) for initial filtering; **Tier 2**—full phonon analysis (dispersion, PDOS) for promising candidates; **Tier 3**—thermodynamic integration, anharmonic properties for final validation. Reduces effective cost by 10–100 $\times$ versus uniform full analysis.

6.4.3 Machine time versus human time in potential development Hidden costs: installation, interface integration, parameter tuning, troubleshooting. Classical potentials: low machine time, high human time (parameterization, validation); uMLIPs: higher machine time, lower human time (pre-trained, standardized interfaces). Total cost optimization considers both components for research group resource planning.

7. Thermodynamic Connections from Phonon Spectra

7.1 Harmonic Thermodynamic Properties

7.1.1 Vibrational entropy and its temperature dependence Vibrational entropy $S_{\text{vib}}(T)$ captures disorder from thermal phonon excitation, with low-T behavior dominated by acoustic modes ($\propto T^3$ in Debye model) and high-T approaching classical limit. Accuracy requirements: ~ 0.1 k_B/atom precision for reliable phase stability ranking at 1000 K ([arXiv.org](https://arxiv.org)) . PFT demonstration: 66% error reduction enables quantitative phase diagram prediction.

7.1.2 Helmholtz free energy contributions to phase stability $F_{\text{vib}}(T)$ integrates energy and entropy contributions, with competing phases typically separated by 10–100 meV/atom in total free energy. Phonon errors of 1–10 meV/atom in F_{vib} can qualitatively alter stability rankings, particularly near phase boundaries. Quasiharmonic approximation extends harmonic phonons to finite temperature through volume-dependent frequencies.

7.1.3 Heat capacity predictions and Debye model comparisons $C_V(T)$ provides direct experimental validation pathway through specific heat measurements. Debye model comparison: deviations indicate spectral complexity beyond simple acoustic continuum. Accurate potentials reproduce: T^3 low-T behavior, smooth approach to Dulong-Petit limit, any intermediate anomalies (Schottky peaks, phase transitions).

7.2 Thermal Transport Properties

7.2.1 Phonon group velocities and lifetimes from harmonic spectra Group velocities $v_g = \partial \omega / \partial q$ from harmonic dispersion; lifetimes τ from anharmonic scattering (third-order force constants). Peierls-Boltzmann transport equation combines these for thermal conductivity κ . Harmonic-only predictions (e.g., Cahill-Pohl amorphous limit) provide bounds; full anharmonic treatment required for quantitative accuracy.

7.2.2 Anharmonic force constant extraction for thermal conductivity Third-order force constants $\Phi^{(3)}$ require larger supercells and more displacements than harmonic $\Phi^{(2)}$. Computational cost: $\sim 10\times$ harmonic for direct finite differences; machine learning acceleration (hiPhive, ALAMODE) reduces through sparse sensing and compressed sensing (chris-marianetti.org) . PFT finding: harmonic Hessian training improves anharmonic property prediction suggests curvature accuracy transfers to higher-order derivatives.

7.2.3 Machine learning acceleration of anharmonic phonon calculations ML approaches: direct prediction from harmonic properties, accelerated $\Phi^*(3)$ construction, full neural network surrogate for . Thermal conductivity benchmark: best models achieve $\sim 0.2 \log(W/m\cdot K)$ MAE, with fine-tuned EquiformerV2(FT) at 0.174 (Source) . Challenge: anharmonicity more sensitive to potential details than harmonic properties; larger training data requirements.

7.3 Phase Stability and Transformations

7.3.1 Finite-temperature free energy surfaces Harmonic $F_{\text{vib}}(T)$ provides zeroth-order finite-temperature stability. Quasiharmonic extension: $F(V,T) = E_{\text{static}}(V) + F_{\text{vib}}(V,T)$, with volume-dependent phonons capturing thermal expansion. Self-consistent phonon theory: iterative renormalization for strongly anharmonic systems where harmonic approximation fails.

7.3.2 Dynamical stability boundaries in composition-temperature space Phonon imaginary modes indicate mechanical instability; temperature-dependent renormalization can stabilize or destabilize structures. Phase diagram construction requires $F(V,T)$ evaluation across competing structures, with phonon errors propagating to boundary uncertainties. High-throughput phonon screening enables efficient stability mapping.

7.3.3 Soft mode-driven structural transitions Soft modes ($\omega \rightarrow 0$) at specific q-points drive second-order phase transitions (ferroelectrics, antiferrodistortive). Accurate soft mode description requires: precise curvature near instability, proper treatment of long-range interactions (dipole-dipole for ferroelectrics), anharmonic stabilization at finite T. Benchmark challenge: small errors can qualitatively change predicted transition temperatures.

7.4 Experimental Validation Pathways

7.4.1 Inelastic neutron scattering spectrum comparison INS provides complete dispersion mapping for single-crystal samples, enabling direct (q, ω) comparison. Challenges: sample quality, resolution limitations, background subtraction. Recent advances: time-of-flight spectrometers, single-crystal mapping for complex materials (Emergent Mind) . GLIM opportunity: prioritize INS-characterized materials for experimental anchor points.

7.4.2 Raman and infrared spectroscopy benchmarks Optical spectroscopies probe zone-center ($q=0$) modes with high frequency precision ($<1 \text{ cm}^{-1}$). Raman: symmetric polarization derivatives; IR: dipole moment derivatives. Selection rules enable mode symmetry assignment. Comparison requires: accurate eigenvectors (not just frequencies), proper LO-TO splitting treatment.

7.4.3 Specific heat and thermal expansion measurements Specific heat $C_P(T)$ integrates all vibrational modes with thermal weighting; thermal expansion $\alpha(T)$ probes mode Grüneisen parameters. Both sensitive to low-frequency modes often challenging for potentials. Combined fitting of multiple properties provides stronger validation than single observable.

8. Machine Learning Potential Phonon Benchmarks: State of the Art

8.1 Large-Scale Benchmarking Initiatives

8.1.1 MatterSim benchmark: >10,000 materials, seven uMLIPs Pioneering scale: >10,000 materials with DFT-level accuracy demonstration. Key finding: phonon errors smaller than PBE-PBEsol functional differences establishes achievable target ([Nature](#)) . Limitation: single-potential focus (MatterSim-v1) for stability screening; multi-potential comparison in original benchmark limited to seven models.

8.1.2 PhononBench: 108,843 AI-generated structures for stability assessment Unprecedented scale: 108,843 structures with MatterSim-v1 phonon evaluation. Key result: only 25.83% dynamically stable, with top generative model (MatterGen) at 41.0% ([Source](#)) . Implication: AI structure generation far from solving stability problem; reliable potential evaluation critical for filtering. Open release: complete dataset and workflows enable community extension ([Nature](#)) .

8.1.3 CHIPS-FF: Comprehensive evaluation across multiple property classes Multi-property focus: phonons, elastic constants, formation energies, surfaces, defects. Key innovation: displacement-dependent phonon analysis revealing ORB/OMat instability at small displacements ([arXiv.org](#)) . Finding: MACE, MatterSim, SevenNet consistent across displacements; direct force output architectures problematic. Efficiency tracking: explicit timing and scaling analysis guides practical selection.

8.2 Performance Trends and Insights

8.2.1 Achieving DFT-level accuracy for harmonic phonon properties Consensus achievement: best uMLIPs reach DFT-level harmonic phonon accuracy with 2–10 meV typical errors. Fine-tuned variants <2 meV approach reference data quality limits. Key enablers: large-scale diverse training data, expressive architectures (equivariant message passing, transformers), proper loss function weighting.

8.2.2 Residual errors: systematic versus random components Systematic errors: consistent over/under-binding (softening/stiffening), functional mismatch (PBE training vs. optB88 reference), missing physics (dispersion, magnetism). Random errors: material-specific failures, numerical noise, incomplete training coverage. Diagnostic value: systematic errors guide architecture/training improvement; random errors indicate data scarcity.

8.2.3 Composition and structure-dependent error patterns Emerging patterns: main-group > transition metal > heavy element/rare earth accuracy; cubic > low-symmetry structures; dense > layered (for PBE-trained) > molecular crystals. Van der Waals exception: ALIGNN-FF (JARVIS-trained) excels. Failure mode clustering: specific element combinations (e.g., H with heavy elements), unusual coordination (e.g., square planar in oxides), strong correlation (e.g., late transition metal oxides).

8.3 Fine-Tuning and Domain Adaptation

8.3.1 Phonon-targeted training strategies PFT methodology: direct Hessian supervision with stochastic column sampling for scalability. Co-training with original data prevents catastrophic forgetting. Results: 55% average phonon property improvement, state-of-the-art

thermodynamic and transport predictions ([Source](#)) . Cost-effectiveness: modest training investment (60–140 GPU-hours) for substantial accuracy gains.

8.3.2 Active learning for thermodynamic property optimization Uncertainty-guided sampling: identify materials where potential most uncertain, DFT evaluation, iterative retraining. Efficiency: maximize information per DFT calculation. Application: specialized potentials for targeted chemical spaces (e.g., battery electrolytes, catalysts).

8.3.3 Multi-fidelity training with experimental data integration Emerging direction: incorporate experimental phonon frequencies, thermodynamic data as additional training constraints. Challenges: experimental uncertainty, sample quality variation, temperature effects. Potential benefit: reduce functional bias, improve absolute accuracy, enhance trust for experimental comparison.

8.4 Emerging Challenges and Opportunities

8.4.1 Anharmonicity and temperature-dependent phonon renormalization Beyond harmonic approximation: self-consistent phonons, molecular dynamics with spectral analysis, explicit anharmonic force constants. Current status: ML acceleration emerging but not yet standard; accuracy validation limited. Opportunity: benchmark anharmonic predictions with temperature-dependent experimental data.

8.4.2 Defect and interface phonon modeling Point defects, dislocations, grain boundaries: local symmetry breaking, resonant modes, phonon scattering. Challenge: large supercells, finite-size effects, convergence with defect concentration. ML potential advantage: efficient large-scale simulation enables statistical sampling.

8.4.3 Real-time phonon calculations for inelastic scattering analysis INS/IXS data interpretation: immediate phonon prediction for structure refinement, mode assignment, anomaly detection. Requirement: sub-second phonon calculation for interactive analysis. Efficient uMLIPs approaching this threshold for moderate system sizes.

9. Implementation Recommendations for the GLIM Project

9.1 Standardized Workflow Design

9.1.1 Consistent geometry optimization protocols across all 23 potentials Fixed-lattice FIRE relaxation with 0.005 eV/Å convergence for primary phonon evaluation; optional self-relaxed comparison for integrated assessment. Document: convergence success rates, iteration counts, final force distributions. Flag: materials with convergence failures for separate analysis.

9.1.2 Automated Phonopy integration with ASE-based potential interfaces Standardized ASE calculator interface enabling seamless potential swapping. Phonopy configuration: supercell 12 Å, displacement 0.03 Å (with convergence verification subset), symmetry exploitation, Fourier interpolation to 1000 points/Å³. Quality assurance: acoustic sum rule satisfaction, symmetry preservation, convergence with respect to sampling density.

9.1.3 Quality assurance and reproducibility checks Automated validation: reference DFT phonon recalculation for random subset (e.g., 1% = 120 materials) to verify workflow fidelity. Cross-check: independent implementation (e.g., hiPhive, ALAMODE) for discrepancy identification. Documentation: complete parameter logging, software versions, random seeds.

9.2 Error Metric Selection and Reporting

9.2.1 Tiered metrics: essential, recommended, and extended benchmarks

Tier	Metrics	Purpose	Computational Cost
Essential	_max MAE, stability rate, PDOS Pearson r	Rapid screening, global ranking	Low ($\sim 1\times$ base)
Recommended	Full dispersion MAE/RMSE, band-resolved errors, S_vib/F/C_V at 300 K	Detailed accuracy assessment	Medium ($\sim 2\text{--}3\times$ base)
Extended	Temperature-dependent thermodynamic properties, anharmonic , mode symmetry success	Specialized applications, research	High ($\sim 5\text{--}10\times$ base)

Table 11: Tiered benchmark metrics with purpose and cost guidance.

9.2.2 Visualization strategies for multi-potential comparison Interactive dashboards: dispersion overlay plots, error distribution histograms, composition-property scatter. Static figures: periodic table heatmaps of element-specific errors, performance-ranking tables with confidence intervals. Specialized: soft mode identification, stability boundary maps in composition space.

9.2.3 Statistical significance and confidence interval estimation Bootstrap resampling for MAE confidence intervals given finite sample size. Stratified analysis: by crystal system, by chemistry (binary/ternary/quaternary), by bonding type. Multiple comparison correction for 23-potential ranking (e.g., Tukey HSD). Report: point estimates, 95% CIs, sample sizes for all metrics.

9.3 Computational Resource Planning

9.3.1 Cost estimation for $12,000 \times 23$ phonon calculations Base estimate: ~ 10 GPU-hours for complete benchmark with efficient implementation (assuming ~ 1 minute per material per potential on modern GPU, parallel efficiency $\sim 80\%$). Contingency: $+50\%$ for convergence failures, restarts, extended analysis. Total: $\sim 1.5 \times 10$ GPU-hours or equivalent CPU-GPU mixed resources.

9.3.2 Prioritization strategies for material and potential subsets Phase 1: all 23 potentials on 1000-material representative subset for statistical power and initial ranking. Phase 2: top 10 potentials on full 12,000 for definitive comparison. Phase 3: top 3–5 potentials on extended analysis (temperature-dependence, anharmonicity). Adaptive: community feedback and preliminary results inform prioritization adjustments.

9.3.3 Incremental benchmark expansion and community feedback integration Quarterly releases: preliminary results with clear uncertainty quantification. Feedback mechanisms: issue tracking, discussion forums, direct collaborator input. Protocol refinement: incorporate lessons from early phases into later execution. Final release: complete dataset, analysis code, reproducibility package.

9.4 Integration with Broader Ecosystems

9.4.1 JARVIS-Leaderboard contribution and interoperability Automated submission: CHIPS-FF-style integration for seamless result contribution. Metric alignment: adopt JARVIS-Leaderboard standard metrics where consistent with project goals. Cross-referencing: link to existing potential entries for longitudinal tracking. Update policy: revise entries as improved potentials or reference data become available.

9.4.2 Open data and code release for reproducibility Data release: structures, phonon frequencies, PDOS, thermodynamic properties for all 276,000 calculations (compressed, with clear documentation). Code release: complete workflow from structure input to metric calculation, containerized for portability. License: permissive open source (e.g., MIT, Apache-2.0) to maximize reuse.

9.4.3 Community-driven benchmark evolution and potential improvement Collaborative governance: steering committee with academic, national lab, industry representation. Update cycles: annual major releases with new potentials, expanded materials, refined metrics. Specialized benchmarks: spin-off initiatives for specific applications (e.g., battery materials, nuclear fuels, quantum materials) building on GLIM infrastructure. Ultimate goal: living benchmark that continuously improves with community contribution, enabling sustained progress in interatomic potential development.